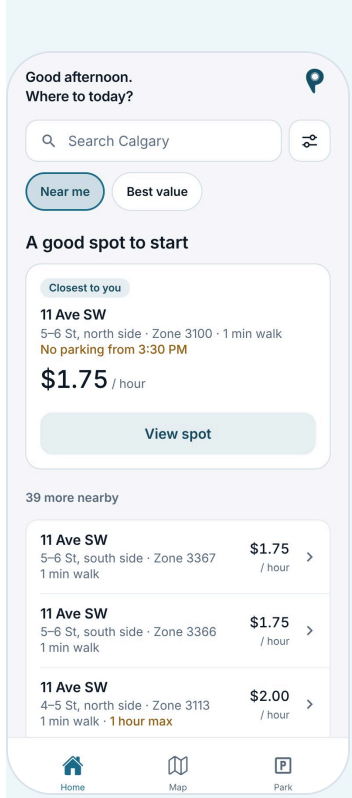
CASE STUDY · EHRL BALQUIN · 2025 TO 2026

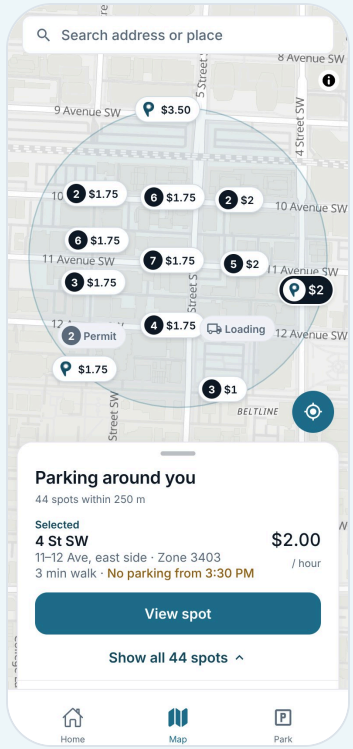
ParkAid

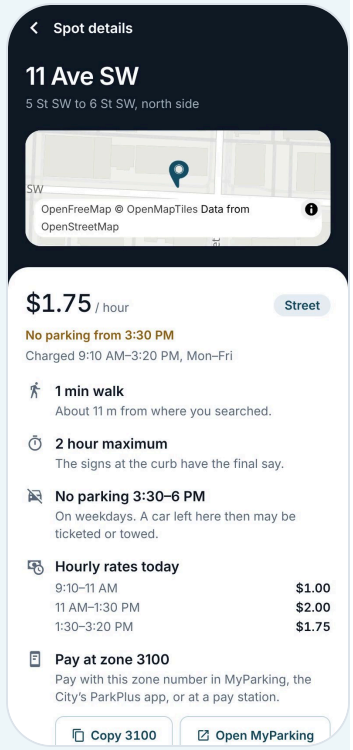
Street parking in downtown Calgary, with the City's real rates, time limits and rush-hour rules.

- Capstone team of five, then my own follow-up work with AI coding help
- React Native (Expo): iOS, Android and web from one codebase
- SAIT capstone, then September 2025 to November 2026

Source code · Demo recordings · Figma file · LinkedIn







THE ROUTE THROUGH THIS CASE STUDY

Overview	Problem	Process	Design	Build	Results
p. 02	p. 03	p. 04	p. 05–07	p. 08	p. 09–10

OVERVIEW

Real rates and rules, in the order a driver needs them

PROBLEM

The capstone app guessed prices from zone codes, showed raw City addresses and listed the same block again and again.

WHAT I DID

Rebuilt the data path and the screens around four driver questions, then ran three rounds of critique and fixes.

RESULT

The City’s own rates and rush-hour rules, a map that reads at a glance, and an honest payment flow.

23.5%

of downtown street rows were repeats (277 of 1,178). Now each comes back once.

45 → 15

pins on the opening map view. 44 overlapped; 4 touch now.

170

automated tests passing: 152 in the app, 18 in the backend.

4.74 : 1

contrast for grey text on a selected row, up from 4.23. AA asks for 4.5.

STACK Expo SDK 54 · React Native 0.81 · React Navigation · react-native-maps · MapLibre GL JS · Node + Express · Postgres + PostGIS · Supabase · Render · Jest

THE PROBLEM

The City's data was right. The capstone app read it loosely.

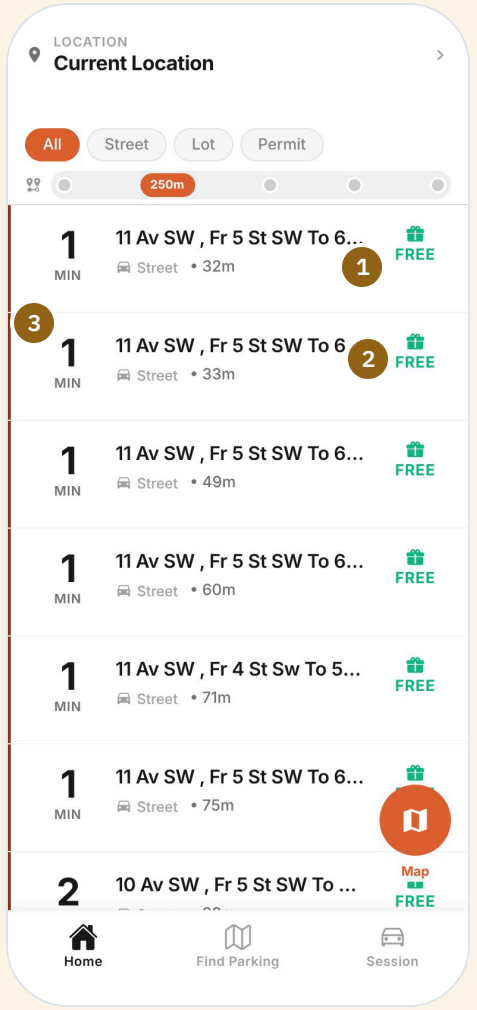
FOUR ANSWERS A DRIVER NEEDS BEFORE LEAVING THE CAR

- **Can I park here now?**
- **What does it cost right now?**
- **Until when?**
- **Which zone do I pay?**

WHAT THE CAPSTONE APP SHOWED, OCTOBER 2025

1. **Raw City addresses**
   "11 Av SW , Fr 5 St SW To 6..."
2. **FREE on a charged block**
   Prices came from a zone-code lookup. Zone 1684 showed $6.00; the City charges $2.00 to $2.25.
3. **The same block, again and again**
   One stretch of curb, split into segments, listed once per segment.

Also: a static map image on the web, and a demo session where you picked the rate yourself.



Capstone code, run with today's City records. Its pricing logic re-created from the old backend.

PROCESS

From a team capstone to work I kept improving

	Capstone	Sep 2025	Mar 2026	May 2026	Oct 2026	Nov 2026
MILESTONES		Code moves to this repo		Backend live, renamed ParkAid	3 critique rounds, 10 fixes	Recordings, Figma, this case study
TEAM OF FIVE	Base app					
ME		Direction and product calls · 27 commits				
AI AGENTS			Claude Code · 101 commits from March 2026			

WHO WROTE THE CODE, TO OCT 28, 2026

101 by Claude Code | **27 mine**

27 commits are under my own accounts, a few of them from an OpenAI Codex branch (April 2026). The team's capstone history isn't in this repo.

HOW EACH ROUND WORKED

Capture
Run the real build with City data

→

Critique
Heuristics, contrast, keyboard, screen sizes

→

Decide
I chose what to fix first

→

Fix
One step per commit

→

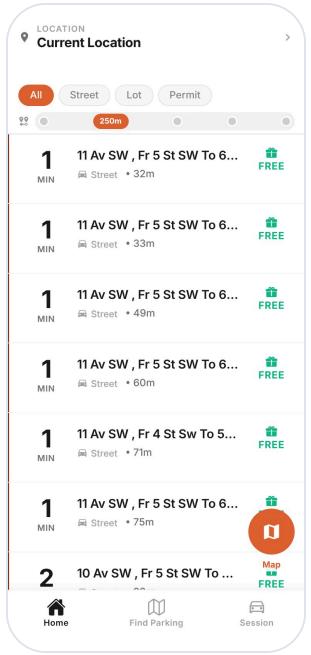
Verify
Jest, 3 screen widths, Android bundle

DESIGN · 1 OF 3

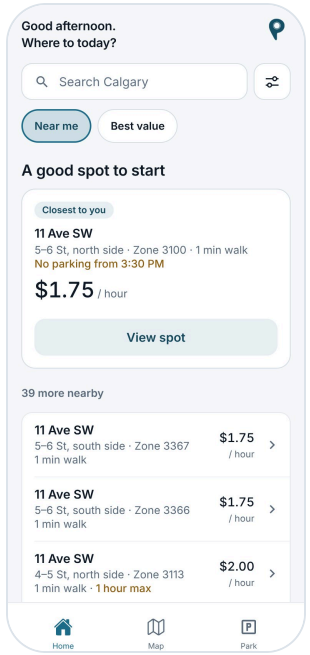
Same City data, clearer answers

Home

BEFORE



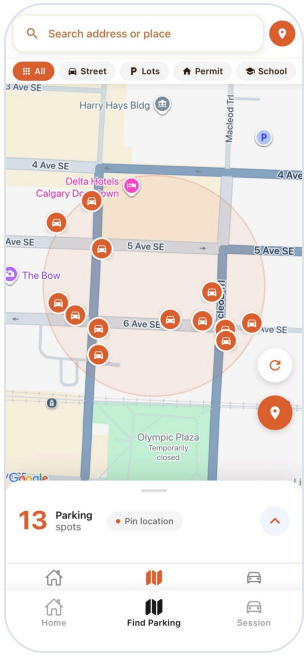
AFTER



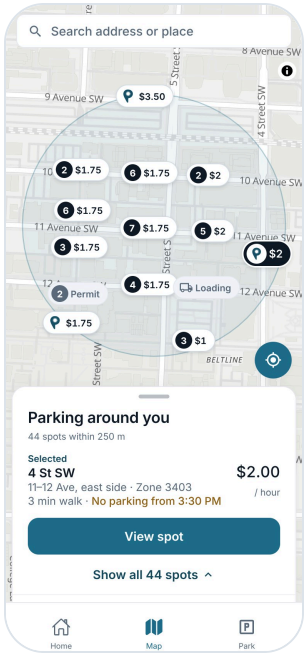
- Block, side and zone
- Rate now, not FREE
- Rush-hour warning

Map

BEFORE



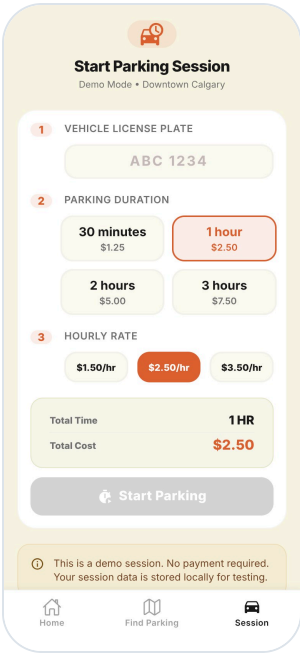
AFTER



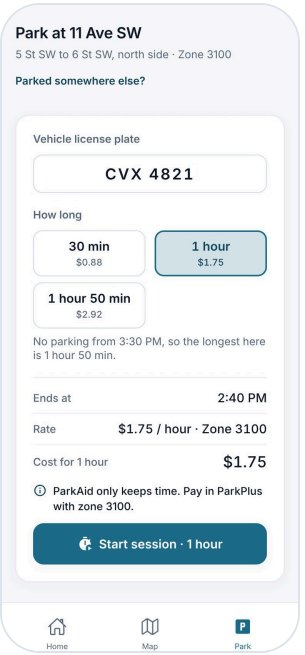
- Real map on the web
- Grouped pins
- Rates on the pins

Park

BEFORE



AFTER



- City rate, not picked
- End time and cost first
- Pay in ParkPlus by zone

DESIGN · 2 OF 3

Five decisions, shown in the app

DECISION 1

Show the rate for now, and for later

Hourly rates today

9:10–11 AM	$1.00
11 AM–1:30 PM	$2.00
1:30–3:20 PM	$1.75

One price is wrong most of the day. Here: $1.00, then $2.00, then $1.75.

DECISION 2

Treat rush hour as a rule

30 min $0.88 | 50 min $1.17

No parking from 3:30 PM, so the longest here is 50 min.

Lengths stop before the tow-away window. No Start inside it.

DECISION 3

Be plain about money

ParkAid only keeps time. Pay in ParkPlus with zone 3100.

Copy 3100 | Open MyParking

ParkAid times the stay. MyParking, the City's app, takes the payment.

DECISION 4

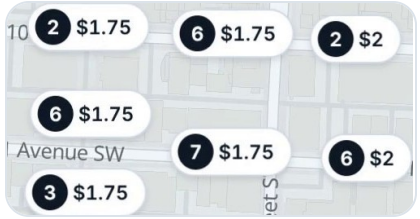
Name blocks the way signs do

11 Ave SW
5–6 St, south side · Zone 3367
1 min walk
$1.75 / hour

Block, side and zone, as a driver reads them at the curb.

DECISION 5

Group pins that would overlap



Count and lowest rate on one pin. A tap zooms in.

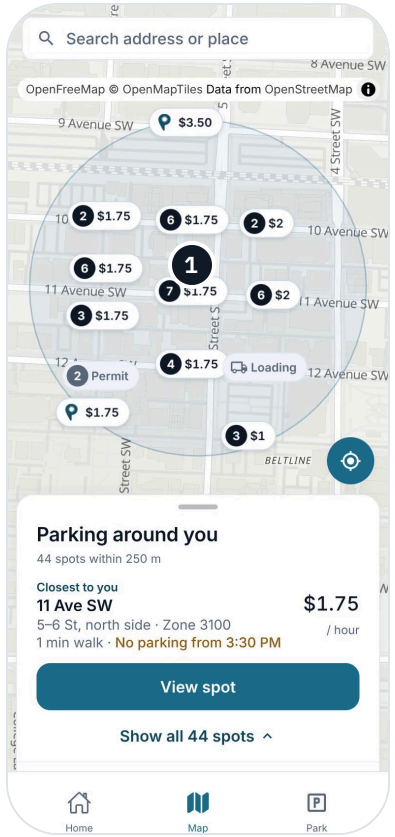
THE RULE UNDER ALL FIVE

> "The signs at the curb have the final say."

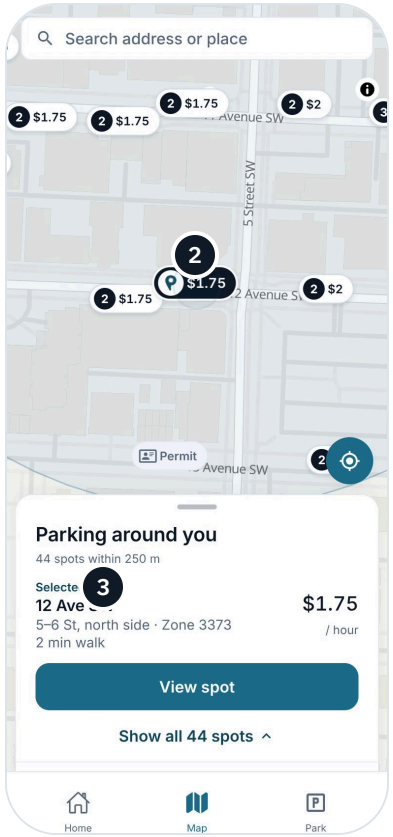
ParkAid's own words, on every spot. The app shows the City's data and says where it stops.

DESIGN · 3 OF 3

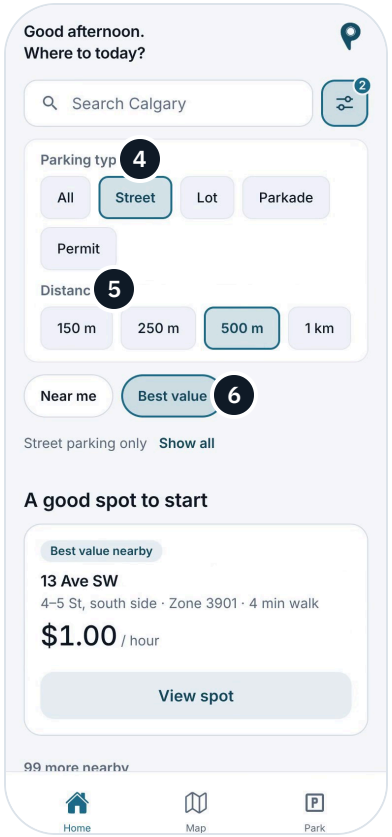
Compare spots without tapping every pin



OPENING VIEW



AFTER A TAP



FILTERS

1. **Group pin** · Count and lowest rate
2. **Tap a group** · Zooms until spots part; the chosen pin turns dark
3. **Sheet** · Follows the selected spot
4. **Type** · Street, lot, parkade, permit
5. **Distance** · 150 m to 1 km
6. **Sort** · Near me by walk, Best value by rate now

THE GROUPING RULE

Pins closer than a group pill (70 × 34 px at the current zoom) share one, led by the best-ranked spot. The same code runs on Google Maps and MapLibre.

```
// utils/mapGroups.js, simplified
for (const spot of spots) {   // best first
  const p = worldPoint(spot, worldSize);
  const near = groups.find((g) =>
    Math.abs(g.x - p.x) < PIN_WIDTH &&
    Math.abs(g.y - p.y) < PIN_HEIGHT);
  if (near) near.spots.push(spot);
  else groups.push({ ...p, lead: spot, spots: [spot] });
}
```

BUILD

One codebase, three platforms, two map engines

City open data
Zones, rates, restrictions

→

Import script
Four datasets, one table

→

Postgres + PostGIS
Supabase; search in SQL

→

Express API
Render; merges segments

→

iOS and Android
react-native-maps (Google)

Web
MapLibre GL JS, OpenStreetMap

MY FRONT-END RESPONSIBILITIES

Screens
Six, with React Navigation

Components
Rows, pins, meta line, actions

Rules
Rates, rush hour, ranking: tested functions

Platforms
A web map file, an Android badge

State
Session context, hooks, offline cache

Access
Labels, Reduce Motion, keyboard

THE HARD PARTS

Rates change by the hour
Rate bands parsed into time ranges; cost counts charged hours only

Rush-hour windows
Checked against now and against the end of a planned stay

Repeated segments
277 of 1,178 street rows merged in the API

Android marker limit
Markers draw into 100 × 100 px; Android gets a compact badge

RESULTS · 1 OF 2

What was checked, what improved, what's still open

CHECKED

Unit tests	152 app, 18 backend: all pass
Real data	1,672 City records near 11 Ave SW
Contrast	4.23 to 4.74 : 1 on selected rows
Keyboard (web)	Pins skipped; search is first
Small phone	360 × 640: actions now on screen
Builds	Web at 360, 390, 1280 px; Android

NOT DONE YET

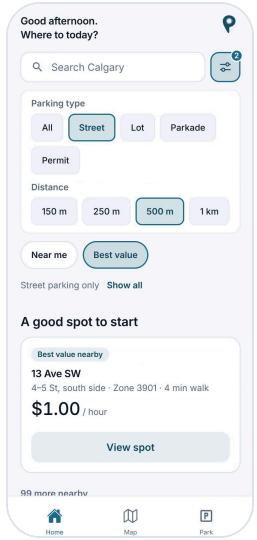
- Usability sessions with drivers
- Testing on a physical phone
- Database test suites and query timing
- The new import on the live database

WHAT IMPROVED

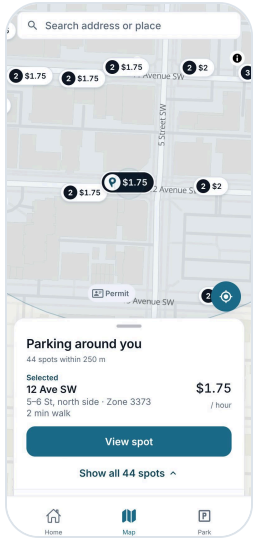
Price	~~Zone-code guess~~	→	**The City's rate for the hour**
Rush hour	~~Not shown~~	→	**Warned, capped, no Start**
Rows	~~Raw address, repeats~~	→	**Block, side, zone, once**
Map on web	~~Static image~~	→	**Real map, grouped pins**
Backend	~~Ran on a laptop~~	→	**Deployed on Render**

RESULTS · 2 OF 2

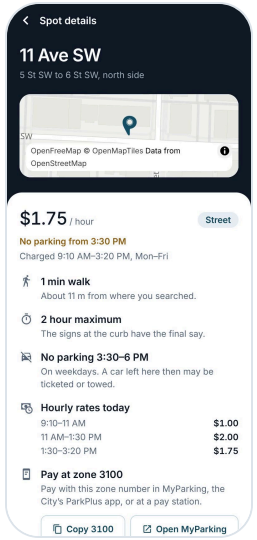
See it working



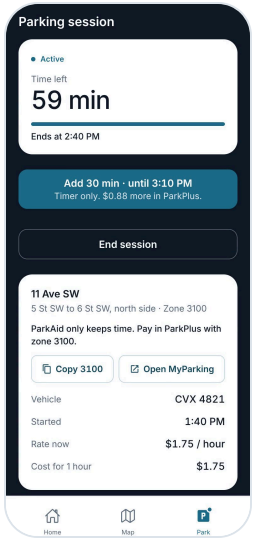
01 Finding parking



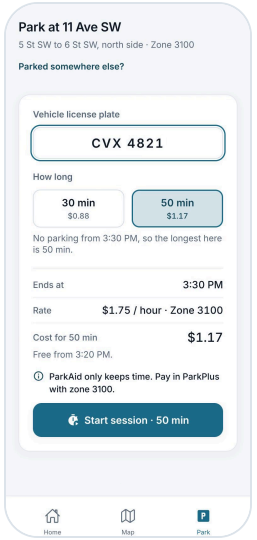
02 The map



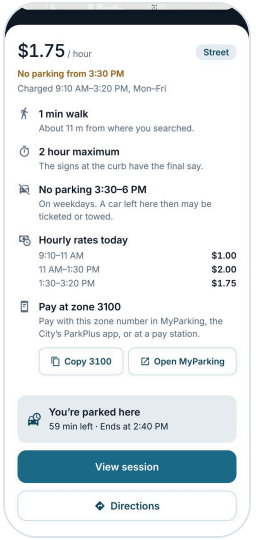
03 Spot details



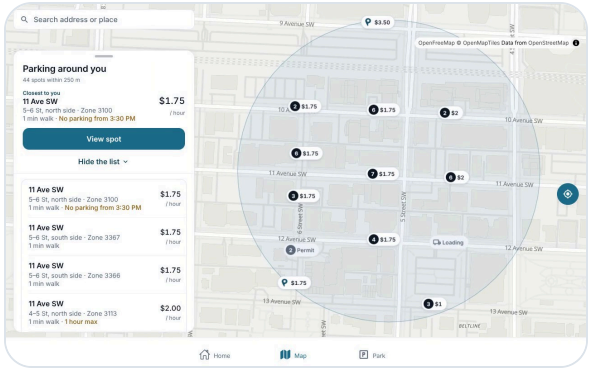
04 Parking



05 Rush hour



06 Parked here



07 On a laptop

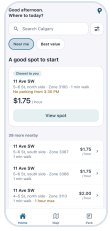
Recorded from the web build with the City's records, at a fixed time. A blue circle marks each tap.

Source code

github.com/ginesbal/parkaid

Demo

Seven recordings, MP4 and GIF



Figma file

Rebuilt from the code: 61 variables, 26 icons, 11 components, 5 screens. Session screens and links still to add.

Made with AI help

Claude Code wrote most of the code from March 2026, plus the recordings, the Figma file and this document's draft. I set the direction and made the calls.